

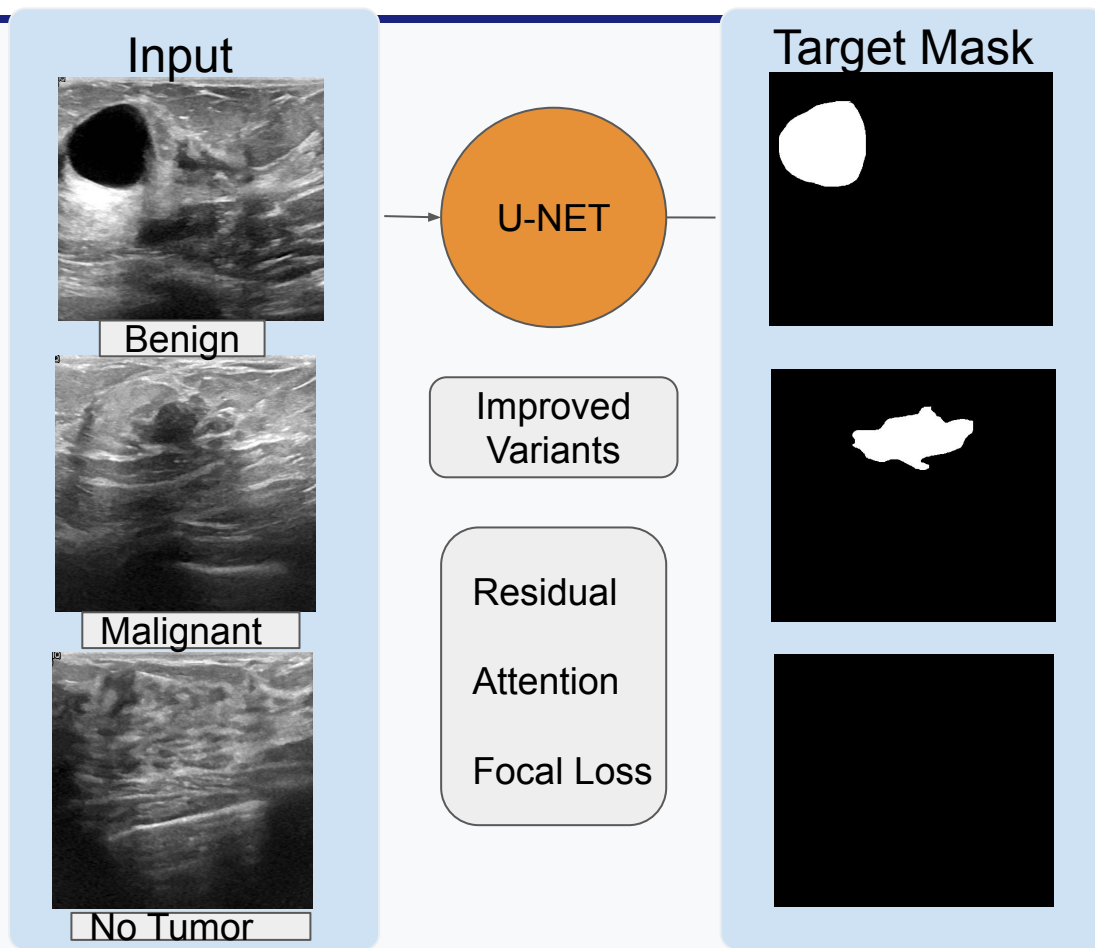
Breast Ultrasound Tumor Segmentation

Deep Learning Analysis with U-Net Variants

GROUP MEMBERS

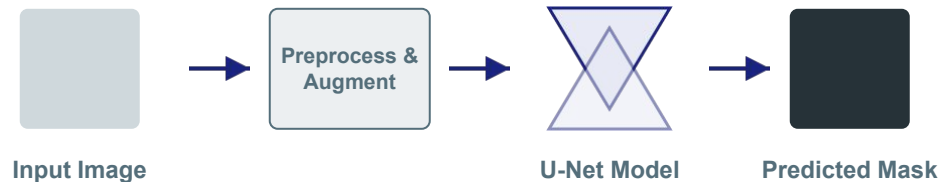
**Joseph
&
Tenzin**

Breast Ultrasound Tumor Segmentation with U-Net Variants

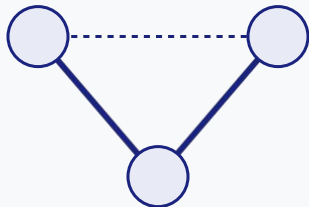


Data and Methods: Pipeline & Model Variants

Segmentation Pipeline



U-Net Structural Logic



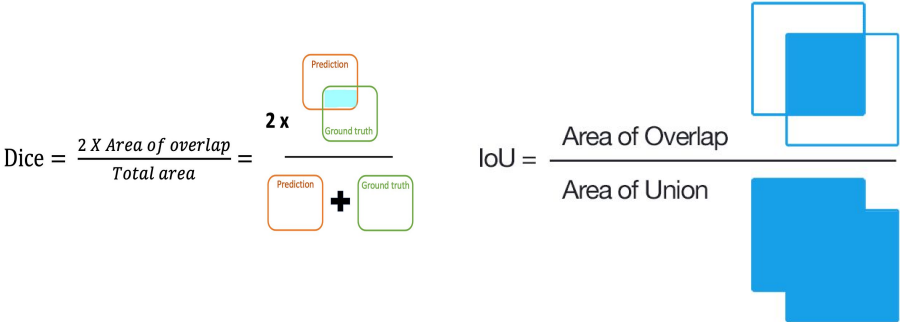
Encoder-Decoder Structure

Skip connections bridge spatial info from encoder to decoder to refine localization.

Results: Loss Function Optimization Yields Best Performance

Comparative Performance Metrics

Variant	Accuracy	IoU	Dice
Baseline Control	0.942	0.482	0.565
Attention	0.942	0.518	0.584
Residual	0.951	0.244	0.661
Swish	0.949	0.618	0.692
BFC (α=0.7)	0.953	0.644	0.727
Baseline (BS=16)	0.955	0.645	0.715
BFC (α = 0.7) + Swish + (BS=16)	0.948	0.609	0.679



Best Performing Model Details

VARIANT

Binary Focal Crossentropy (α=0.7)

ACCURACY

0.953

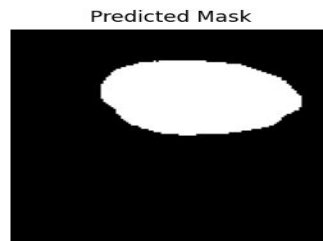
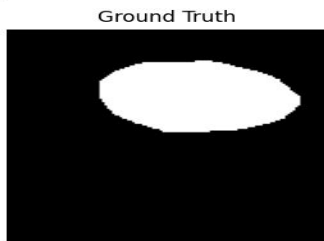
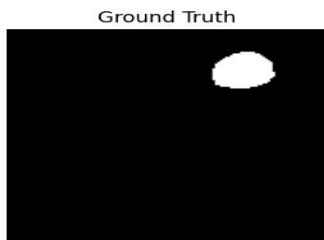
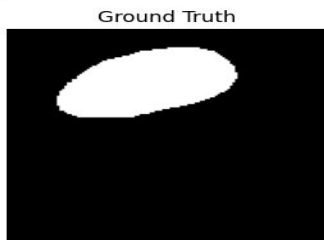
IOU

0.644

DICE

0.727

Representative examples from the best-performing model



Lessons Learned: Beyond Pixel-Level Accuracy



Metric Prioritization

Dice and IoU were more informative than pixel accuracy for tumor segmentation.



Training Design

Loss function and batch size affected performance as much as architecture changes.



Project Limitations

Results were limited by class imbalance, and a simplified evaluation protocol.



Thank You :)